Exercise: Visualizing data by geographical location

Introduction

By now, you have a deeper knowledge of how map visuals can be used effectively in Microsoft Power BI and how they add visual impact to reports. You have explored the essential concepts of map visuals such as **Shape maps**, **Choropleth maps**, **Azure maps** and **Filled maps** and experienced how they are used to generate regional insights and create geo hierarchies from your data.

In this exercise, you can apply this knowledge while creating an enhanced sales report for Adventure Works.

Scenario

Adventure Works sells its products globally, generating a large amount of sales data. As a result, it is important to create more focused summary reports to help regional managers understand the information and predict trends in their own geographical regions. Adio, one of the regional sales managers in the United States, has requested such a report. As a data analyst and experienced report designer, you know that while Power BI offers a range of map visualization options, the **Shape map** visual is the appropriate choice because you will be focusing on data from only one country.

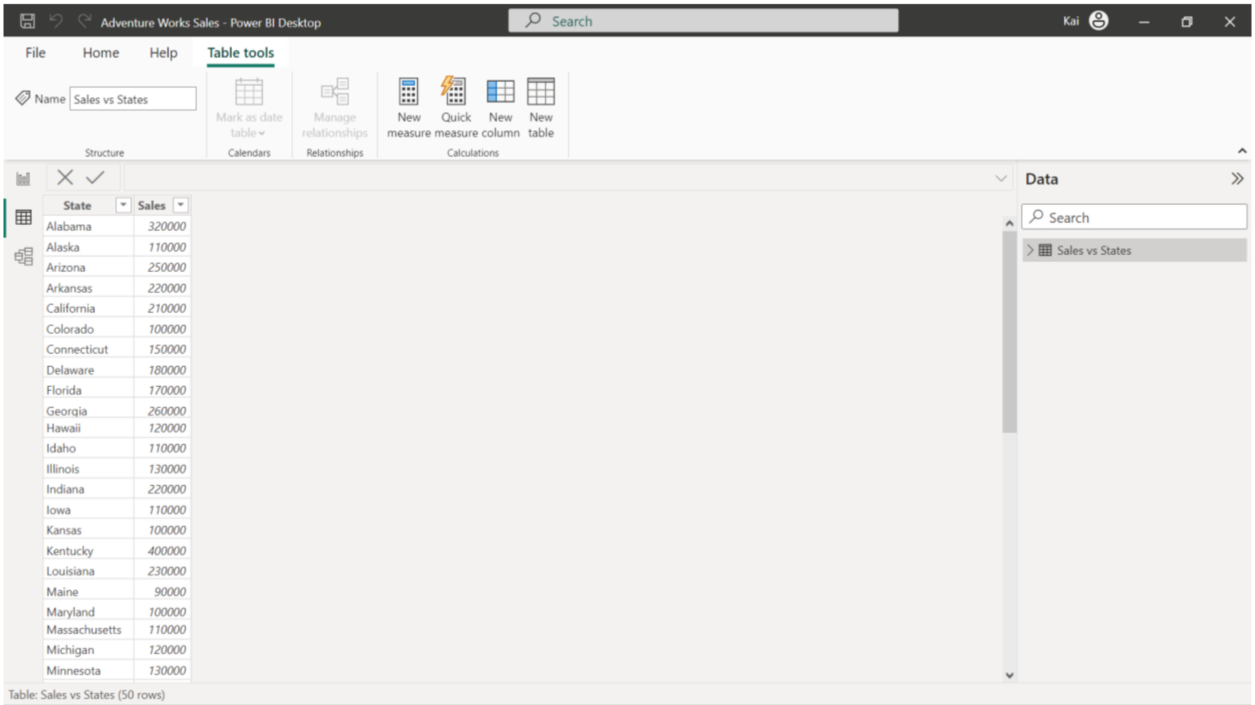
By completing this exercise, you will demonstrate your ability to:

- Review and format the data to be used in the map visualization.
- Create a **Shape map** visual in the Power BI report to display the sales data for specific US states.
- Format and configure the map appropriately using color coding and design perspectives.

Instructions

Step 1: Download the Power BI project File

Download and open the *Adventure Works Visualizing data by geographical location.pbix*. This file contains a single data table **Sales vs States** which contains two data fields: **Sales** and **States**.



Step 2: Review and format the data type

1. Review the data in the two data files. Assess the field categories and types and adjust the format where the data type is not appropriate for use in map visualization.

Tip: You review and change the data type from the **Column Tools** tab in Power BI desktop.

Step 3: Create a Shape map visual to present the sales data

1. Create a **Shape map** visual. In the **Data pane** select and add the appropriate data fields from the data pane to the map visual fields..
2. Resize the map visual to fill the report canvas.

Tip: The **Visualization pane** provides you with all the information you need to build the map visual.

Step 4: Format and configure the map visual

1. Apply the **Accessible city park** theme. Format the color scheme of the map to create a color coding that represents the sales trend across the states.

2. Configure and validate the zoom control of the map so the report users can manually select a specific region in the map.
3. Format the title and tooltip to create an easy-to-understand experience of the visualization.

Tip: You can access the formatting options in the **Visual** and **General** tabs of the **Visualization pane**.

Step 5: Save the Power BI project

1. Save the amended Power BI report locally on your computer.

Conclusion

The **Shape map** visualization that you have successfully created and configured, will display the sales data for the United States in a layered way. It will allow report users such as Adio to view overall sales information while also being able to drill-down to more focused regional analysis.